Assignment 5

Christian Parker
CXP22004

October 19, 2026

1.

$$\begin{aligned} E(I_\alpha, J_\alpha = 1, J_\beta) &= c(I_\alpha - 1)^2 + \sum_{\{\beta:\alpha\sim\beta\}} (1 - J_\beta)^2 + C \\ &= c(I_\alpha - 1)^2 + \sum_{\{\beta:\alpha\sim\beta\}} (J_\beta - 1)^2 + C \end{aligned}$$

$$\begin{aligned} E(I_\alpha, J_\alpha = 0, J_\beta) &= c(I_\alpha - 0)^2 + \sum_{\{\beta:\alpha\sim\beta\}} (0 - J_\beta)^2 + C \\ &= c(I_\alpha)^2 + \sum_{\{\beta:\alpha\sim\beta\}} (J_\beta)^2 + C \end{aligned}$$

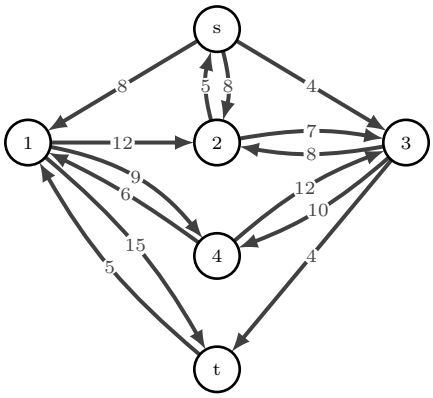
C represents all terms that are independent of J_α

$$\begin{aligned} p &= P(J_\alpha = 1|_\alpha J) \\ &= \frac{P(J_\alpha = 1,\ {}_\alpha J)}{P(J_\alpha = 1,\ {}_\alpha J) + P(J_\alpha = 0,\ {}_\alpha J)} \\ &= \frac{\frac{1}{Z'} e^{\frac{-E(I_\alpha, J_\alpha = 1, J_\beta)}{T}}}{\frac{1}{Z'} e^{\frac{-E(I_\alpha, J_\alpha = 1, J_\beta)}{T}} + \frac{1}{Z'} e^{\frac{-E(I_\alpha, J_\alpha = 0, J_\beta)}{T}}} \\ &= \frac{e^{\frac{-c(I_\alpha - 1)^2 - \sum_{\{\beta:\alpha\sim\beta\}} (J_\beta - 1)^2 - C}{T}}}{e^{\frac{-c(I_\alpha - 1)^2 - \sum_{\{\beta:\alpha\sim\beta\}} (J_\beta - 1)^2 - C}{T}} + e^{\frac{-c(I_\alpha)^2 - \sum_{\{\beta:\alpha\sim\beta\}} (J_\beta)^2 - C}{T}}} \\ &= \frac{e^{\frac{-c(I_\alpha - 1)^2 - \sum_{\{\beta:\alpha\sim\beta\}} (J_\beta - 1)^2}{T}}}{e^{\frac{-c(I_\alpha - 1)^2 - \sum_{\{\beta:\alpha\sim\beta\}} (J_\beta - 1)^2}{T}} + e^{\frac{-c(I_\alpha)^2 - \sum_{\{\beta:\alpha\sim\beta\}} (J_\beta)^2}{T}}} \end{aligned}$$

2. There are two terms in the energy function E, one measures the distance between the image I and the segmentation J, the second measures the smoothness of J. In the simulated annealing algorithm, the distance term is weighted by c. Therefore, for higher c, the distance term has greater influence on J than the smoothness term.

Across the values of c tested, we can see the segmentation transition from noise to an outline of a dog as c increases. At low values of c, when smoothness is weighted highly, the noisiness of the initial uniform distribution dominates. Since the distribution is uniform, the smoothness term is a coin flip rather than pulling the probability distribution towards the most frequent value. At high values of c, the noisiness is overwhelmed by the weight of the distance term. The probability distribution is pulled towards the input image, shrinking the distance between I and J, and aligning the segmentation with the image.

3. (a) Residue network:



(b)

$s \to 1 \to t$

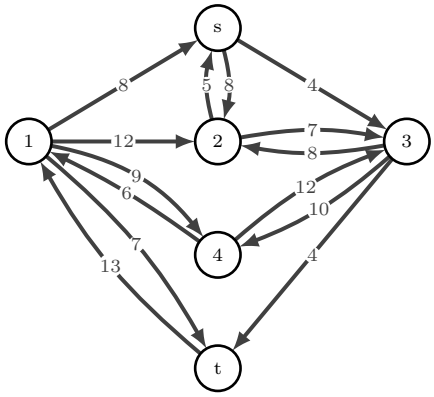
$s \to 2 \to 3 \to t$

$s \to 3 \to t$

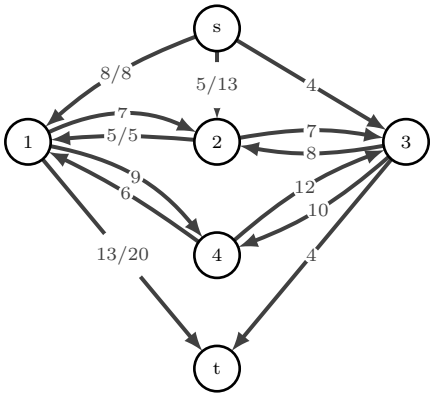
(c)

Augmenting Path: $s \to 1 \to t$

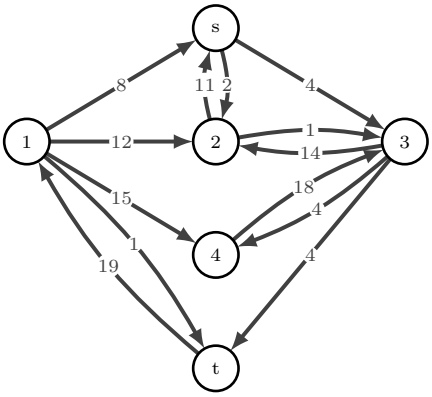
Residue network:



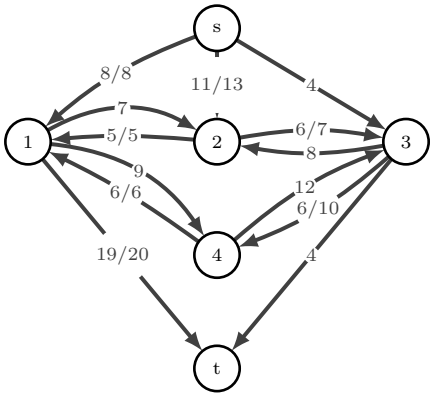
Flow network:



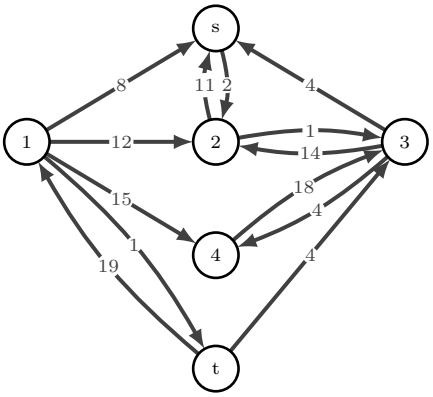
Augmenting Path: $s \to 2 \to 3 \to 4 \to 1 \to t$
Residue network:



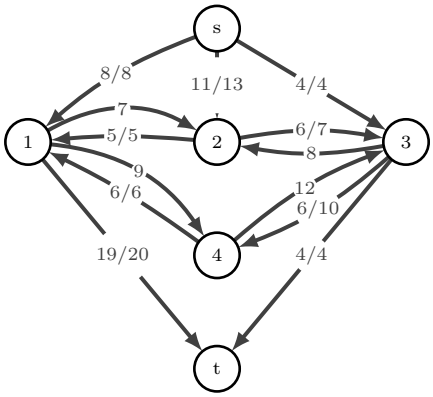
Flow network:



Augmenting Path: $s \to 3 \to t$
Residue network:



Flow network:



(d)

$$\begin{aligned}
\text{Max flow} &= f(s,1) + f(s,2) + f(s,3) \\
&= 8 + 11 + 4 \\
&= 23 \\
\text{Min cut} &= w(s,1) + w(2,1) + w(4,1) + w(3,t) \\
&= 8 + 5 + 6 + 4 \\
&= 23
\end{aligned}$$